PULSEIQ WORKFORCE ANALYTICS

PHASE 2 — PROJECT PROGRESS REPORT

Prepared by: Aiswarya A.
Track: Data Analytics
Reporting Period: October 2026
GitHub Repository: https://github.com/Aiswarya3006/PulseIQ
Release Tag: phase-2
Branch: main

1. EXECUTIVE SUMMARY

Phase 2 of the PulseIQ Workforce Analytics project focuses on transforming workforce and payroll data into meaningful analytical outputs through interactive dashboards and metric-based reporting.

The work covers attendance and leave analysis, attrition analytics, workforce dashboard development, metric/KPI analysis, and budget-versus-actual payroll comparison. The objective is to make key workforce indicators easier to interpret while maintaining traceability to the underlying data.

The Task 10 dashboard and its supporting evidence have been committed and pushed to the project's GitHub repository. The phase-2 tag has also been pushed successfully.

2. TASK DELIVERY SUMMARY

Task	Deliverable	Status	Evidence
Task 6	Attendance and Leave Analysis	Completed	Attendance/leave dashboard screenshot
Task 7	Attrition Analytics	Completed	Attrition dashboard screenshot
Task 8	Workforce Dashboard	Completed	Dashboard overview and filter screenshot
Task 9	Metric/KPI Analysis	Completed	Task 9 output or validation screenshot
Task 10	Budget and KPI Dashboard	Completed	Dashboard, charts, hand verification, and recommendations screenshots

Confirm each task status against the original task requirements and insert the corresponding evidence before submission.

3. IMPLEMENTATION AND EVIDENCE

3.1 Workforce Analytics

- Workforce dashboards track attendance, leave, and attrition using interactive filters.
- Predicted attrition risk supports decisions but does not guarantee outcomes.

3.2 Budget and KPI Analysis

Task 10 compares budgeted payroll with actual employer cost and presents the resulting variance and variance percentage.

The dashboard-level summary is:

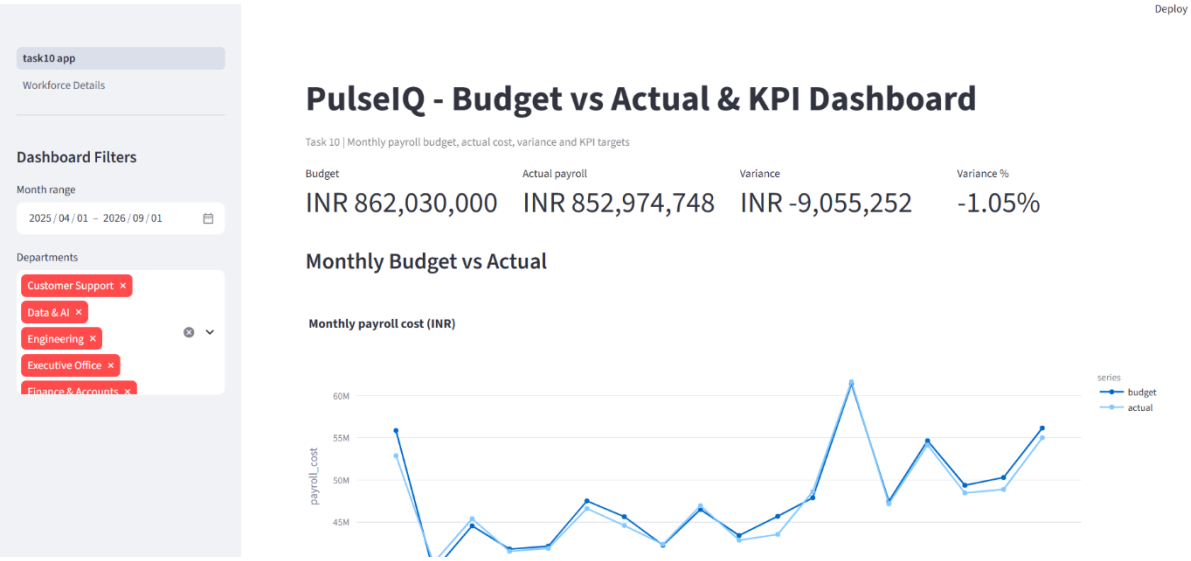
Metric	Reported Value
Budgeted payroll	₹862,030,000
Actual payroll	₹852,974,748
Variance (Actual − Budget)	−₹9,055,252
Variance percentage	−1.05%

A negative variance means actual payroll is below the budget for the displayed reporting scope. This indicates an underspend, but it does not automatically establish a saving; budget assumptions and workforce changes should also be reviewed.

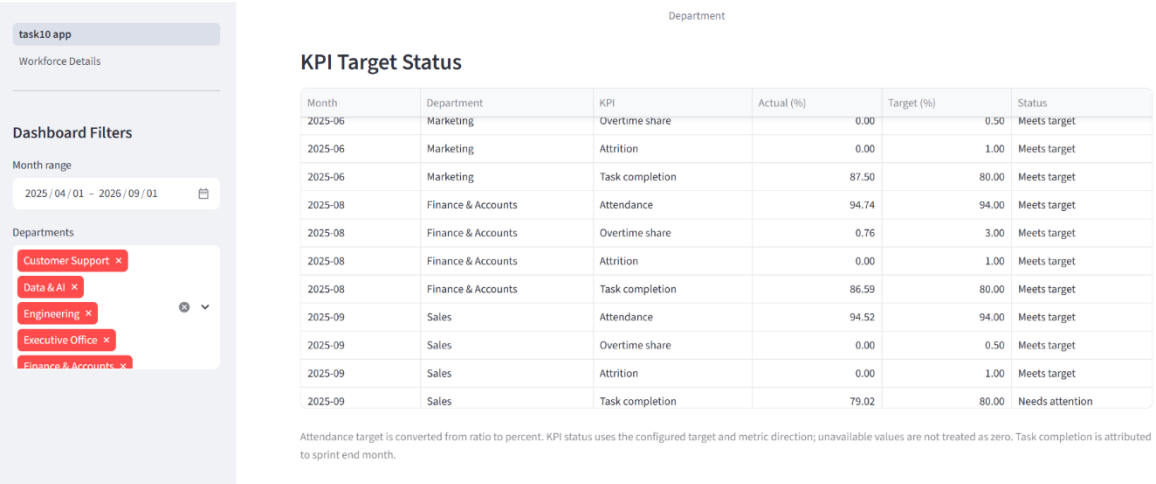
3.3 Repository Evidence

The following Task 10 evidence files are included in the repository are given below

reports/task10_dashboard_overview.png



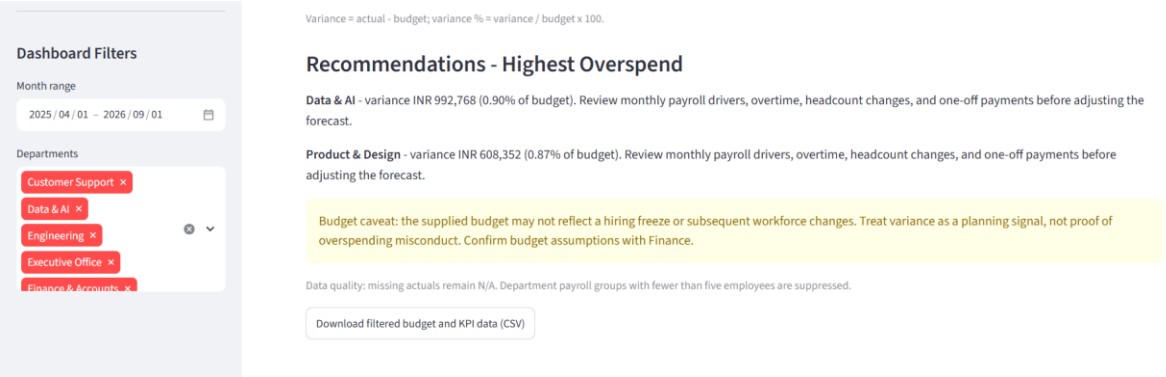
reports/task10_charts_kpi.png



reports/task10_hand_verification.png



reports/task10_recommendations.png



4. METRIC RECONCILIATION AND MANUAL VERIFICATION

The reconciliation process compares budgeted payroll with actual employer cost for the same department and reporting period.

The calculation follows these definitions:

- Variance: Actual payroll − Budgeted payroll
- Variance percentage: (Variance ÷ Budgeted payroll) × 100
- Interpretation: Positive variance indicates an overspend; negative variance indicates an underspend.

Five-Line Data-to-Chart Explanation

1. The monthly department budget is obtained from raw.budget_payroll_dept_month.
2. Actual employer cost is obtained from the main.m09_employer_cost metric view.
3. Both values must refer to the same department and reporting period.
4. Variance and variance percentage are calculated using the formulas above.
5. The independently calculated values are compared with the dashboard output to verify the reported result.

Validation evidence: Insert the hand-verification screenshot showing the selected department, month, source budget, actual cost, and calculated variance. Use the same reporting scope as the displayed dashboard figures.

5. TESTING AND VALIDATION

The Task 10 dashboard was run locally, and supporting screenshots were saved for the overview, KPI charts, manual verification, and recommendations. The implementation and these evidence files were committed and pushed to GitHub.

Only verified checks should be reported as passed. Add any available test command, automated test output, or metric-check result before final submission.

6. CHALLENGES AND RESOLUTIONS

During development, attention was given to making calculated metrics and dashboard outputs easier to interpret. Supporting screenshots were captured to document the implementation and provide evidence for review.

Streamlit deprecation warnings relating to layout parameters were observed during development. These warnings should be distinguished from application execution errors and addressed where appropriate. Final validation should confirm that the dashboard runs correctly and that displayed calculations agree with the underlying data.

7. KEY LEARNINGS

This phase strengthened my understanding of the relationship between raw data, analytical metric views, and dashboard visualisations. I learned how to calculate and interpret budget variance, why comparisons must use consistent reporting periods, and how independent calculations help validate dashboard results. I also learned to present recommendations carefully, distinguishing measurable findings from assumptions that require further investigation.

8. NEXT-PHASE PLAN

The next phase will focus on reviewing feedback, resolving any identified issues, and strengthening the validation and documentation of the project. I will continue maintaining the GitHub repository, ensure that evidence is available for each deliverable, and improve the traceability and clarity of the reported metrics.

9. SUBMISSION INFORMATION

Project: PulseIQ Workforce Analytics
Track: Data Analytics
Phase: Phase 2
Repository: https://github.com/Aiswarya3006/PulseIQ
Git Tag: phase-2
Report File: DataAnalytics_Aiswarya_Phase2_Report.pdf

This report is intended to provide a concise record of the Phase 2 deliverables, implementation evidence, validation approach, lessons learned, and next steps.